Cartilaginous Tissu

1. Introduction

Cartilaginous tissue is a specialized connective tissue of mesenchymal origin (p. 1). It is characterized by a hard but non-mineralized consistency. It consists of specific cells called chondrocytes and an extracellular matrix (ECM) (p. 1).

There are three types of cartilage:Hyaline cartilage ,Fibrocartilage and Elastic cartilage

Localization according to age:

Fetus: Makes up the majority of the skeleton

Child: Importance decreases (e.g., growth plate/epiphyseal cartilage)

Adult: Less widespread (found in articular surfaces, ribs, respiratory tract, ears, nose)

Key Property: Unlike most connective tissues, cartilage lacks vascularization and innervation. It is nourished by a surrounding connective envelope called the perichondrium

2. Histological Structure

Cartilage is composed of cells, fibers, and ground substance

2.1. Cells

These cells are responsible for synthesizing the ground substance and fibers.

Chondroblasts (Young cells): Feature an irregular plasma membrane, a central spherical nucleus, and abundant organelles (Rough ER, Golgi apparatus, mitochondria) for synthesis

Chondrocytes (Mature cells): These are the most mature elements, housed in small cavities called chondroplasts

Chondroclasts: Giant multinucleated cells with a brush border. They are rich in lysosomes and enzymes (like metalloproteinases) required for cartilage resorption

2.2. Ground Substance

Basophilic, homogeneous, and translucent

Rich in water and mineral salts

Composed of sulfated proteoglycans, including chondroitin sulfate, keratan sulfate, and hyaluronic acid

2.3. Fibers

Mainly collagen fibers (Types I, II, or IX) and elastic fibers

Collagen fibers are organized into "baskets" around chondrocytes, forming units known as chondromes

3. Classification of Cartilaginous Tissue

3.1. Hyaline Cartilage

The most common variety in the body

Composition: ECM contains 40% ground substance and Type II collagen fibers (no elastic fibers)

Appearance: Amorphous and homogeneous under a light microscope because microfibrils are too small to be seen

Locations: Nasal fossae, thyroid and cricoid cartilages of the larynx, tracheal rings, and articular surfaces of mobile joints (e.g., the knee)

3.2. Fibrocartilage (Fibrous Cartilage)

An intermediate tissue between dense oriented connective tissue and hyaline cartilage

Distinction: Contains abundant, thick Type I collagen fibers organized in bundles

Locations: Found in mechanical stress zones such as intervertebral discs, articular menisci, pubic symphysis, and the Achilles tendon

3.3. Elastic Cartilage

Characterized by an abundance of elastic fibers forming a dense 3D network

Locations: External ear (pinna), epiglottis, and external auditory canal

4. The Perichondrium

A sheath of connective tissue surrounding the cartilage (except at articular surfaces). It has two layers

Outer Fibrous Layer (Nutritive): Dense connective tissue containing thick collagen fibers, elastic fibers, fibroblasts, and significant vascularization

Inner Cellular Layer (Chondrogenic): Loose connective tissue containing fine collagen fibers (Sharpey's fibers) that penetrate the cartilage matrix

5. Characteristics and Growth

Regeneration: Very low power of regeneration, which further decreases with age

Growth:

Interstitial Growth: Occurs via mitosis of chondrocytes. If they divide in one direction, they form axial isogenic groups; if in various directions, they form coronal isogenic groups

Appositional Growth: New chondroblasts derive from the perichondrium, creating new matrix on the surface .

Nutrition: Provided by blood capillaries in the outer perichondrium or by synovial fluid for articular cartilage